

Course Name: Cyber Security & Forensics - II

Course Code: 05201352

Prerequisite: Basic Knowledge of Networking

Rationale: True Network Security skills for Computer Science Student

Course Learning Objective:

CLOBJ 1	Understand fundamental concepts of network security, including protocols, authentication mechanisms, and their real-world applications.
CLOBJ 2	Identify and analyze software vulnerabilities and malware, understanding their characteristics, propagation methods, and impact on systems.
CLOBJ 3	Explore various vulnerability scanning techniques and defensive tools, enabling detection and mitigation of security threats
CLOBJ 4	Define and explain core concepts of Enterprise Messaging Systems (EMS), IP security, and web security, including protocols and implementation strategies.
CLOBJ 5	Understand different types of cyberattacks, cyber laws, and protective measures, promoting awareness and responsibility in the digital society.
CLOBJ 6	Demonstrate the ability to apply learned security principles to protect personal, organizational, and societal digital assets effectively against cyber threats.

Course Learning Outcomes:

CLO 1	Understand the concepts of Network security and their use.
CLO 2	Understand vulnerabilities in software flaws and concept of malware.
CLO 3	Understand vulnerability scanning and defence tool.
CLO 4	Define the concepts of EMS, IP security and web security
CLO 5	Understand attack, types of cyberattack, cyber laws and how to protect them self and ultimately society from such attacks
CLO 6	After Learning the course, the students shall be able to

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	1	2	-	5	20	20	20	60	30	150

SEE - Semester End Examination, T - Theory, P - Practical

Course Content

W - Weightage (%) , T - Teaching hours

Sr.	Topics	W	T
1	Cyber Security Introduction and Basic requirements Basics of Linux systems kali, Santoku(Mobile Forensics), Ubuntu, Introduction to Virtualization and its types.Introduction on Virtualization Software (VMWare/ VirtualBox). Installation of OS with virtualization Software with ISO file of .vm(vmware file extension), .vhd(Virtual box file extension) files. Cryptography and Steganography Basic Terminology. Protocols, Communication using Symmetric Cryptography. Introduction to One-way	17	8

	Sequence Generators. Introduction to Basic, Intermediate, Advanced and Esoteric Protocols.		
2	Forensics Introduction Chain of Custody, challenges in forensics, Process of investigation, Basics of Law, Understanding Cyber Space, Defining Cyber Laws, Introduction to IT Act 2000, Amendments in IT Act, Goals of Incident Response, Incident Response Plan, Incident Identification	10	5
3	Networking Basics Protocols, Packets: The IP Header, Protocol Layers, ISO/OSI, Encapsulation, IP : IPv4 Addresses, Reserved IP Addresses, IP/Mask and CIDR, IP/Mask CIDR Example, IP/MASK Host Example, Network and Broadcast Addresses, IP Examples, Subnet Calculators, Routing, Routing Table, Routing Table Example, Default Route Example, Routing Metrics, Routing Metrics Example, Checking the Routing Table, TCP and UDP Headers	10	5
4	Firewall and Defense Practical Session: Wireshark, NIC Promiscuous Mode, Configuring Wireshark, The Capture Window, Filtering, Capture Filters, Display Filters	10	5
5	Digital Forensics Introduction and File System Basics of Networking, Registry, MRU Bags, Shall Bags, Introduction to Digital Forensics, Lockard's Principle of Exchange in Digital Forensics, Pre-requisite for setting up Digital Forensic lab and global standards, File types and signature, Architecture and Functioning of memory devices, Retrieval of information, etc., Windows file systems, UNIX /Linux file systems, MAC file systems, Mobile file systems, Network file system, other types of file systems	15	7
6	Log System Analysis of registry in various operating systems, which includes system logs, kernel logs, event logs, ftp/sftp, application, Web Servers/ Proxy logs. (Practical) Digital Forensics Tool Introduction: Working with Autopsy, Using FTK to solve a Cyber Crime, EnCase	10	5
7	Vulnerability Assessment Introduction to Vulnerability Assessment, Life cycle of Vulnerability Assessment, Vulnerability Scanners, Manual Testing, Vulnerability using W3af and Nikto, Nessus, Foot printing of all network scanner like nmap. HTTP Verbs: GET, POST, HEAD, PUT, DELETE, OPTIONS, Using HTTP 1.0 Syntax, Introduction to OWASP TOP 10 Vulnerabilities. Password Attacks - Brute Force Attacks: A Brute Force Algorithm, Brute Force Weaknesses, Dictionary Attacks, Performing a Dictionary Attack, Weaknesses of Dictionary Attacks, Mangling Words, John the Ripper	10	5
8	Malware Analysis Malware Definition and Types, Malware Analysis, Forensic Importance of Malware Analysis, Introduction to different analysis techniques, Malware Behavior, Setting up malware analysis laboratory		
9	Static Analysis Hashing, Finding Strings, PE Files and Headers, Linked Libraries and Functions, Malware analysis in Virtual Machines. Dynamic Analysis: Sandboxes, Running and Monitoring a Malware, Process Monitor, Process Explorer, RegShot, Using Wireshark for Packet Analysis.		

List of Practical

1.	Configuring and installing OS on VMWare or VirtualBox
2.	NMAP: Introduction of NMAP list out some NMAP commands and perform TCP port scanning. Perform various network monitoring task using NMAP commands (at least ten).
3.	WIRESHARK: Introduction to Wireshark and carrying out packet filtering Carrying out source and destination IP address detection using packet capturing in WireShark
4.	W3af: Introduction to w3af and configuring plugins and target on w3af console. 1. Perform various vulnerability scans on different target sites using w3af GUI. 2. SQL injection on a sample website using w3af.
5.	EXTRA: Perform manual SQL injection on a predefined vulnerable website Introduction to Sqlmap and perform SQL injection attack on a sample website.
6.	Introduction and configuration of DVWA. Perform XSS attack on DVWA
7.	Digital Forensics Practical of Autopsy with the case file and try to collect all the data like deleted files, Website browsing history, System information including password, User Details, Attached Devices
8.	Malware Analysis: Follow Practical from Practical Malware Analysis Book